

Ex// Consider the collection of vectors in $[0, 2\pi]$ $\{1, \cos(2x), \sin^2(x)\}$.
This collection is dependent since

$$\left(\frac{1}{2}\right) 1 + \left(-\frac{1}{2}\right) \cos(2x) + (-1) \sin^2(x) = 0$$

due to the double-angle identity in trigonometry.

Ex/ Suppose we had any two differentiable functions ϕ_1 & ϕ_2 . We wish to tell if they are dependent. Thus, we need c_1 and c_2 s.t.

$$c_1 \phi_1 + c_2 \phi_2 = 0.$$

But this alone is not enough to determine c_1 and c_2 . We need another condition involving c_1 & c_2 . If we differentiate both sides of the equation, this gives us the condition we need!

$$c_1 \phi_1 + c_2 \phi_2 = 0$$

$$c_1 \phi_1' + c_2 \phi_2' = 0$$

$$\text{or } \begin{pmatrix} \phi_1 & \phi_2 \\ \phi_1' & \phi_2' \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system has a unique solution of $\vec{c} = \vec{0}$ iff $\phi_1 \phi_2' - \phi_2 \phi_1' \neq 0$
(from Cramer's Rule).

Ex// this also extends for n functions that are $(n-1)$ times differentiable. If the n functions are dependent, then there will be a non-zero $\vec{c}$ s.t. $W\vec{c} = \vec{0}$, where

$$W = \begin{pmatrix} \phi_1 & \phi_2 & \dots & \phi_n \\ \phi_1' & \phi_2' & \dots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n-1)} & \phi_2^{(n-1)} & \dots & \phi_n^{(n-1)} \end{pmatrix}$$

Definition: The Wronskian of n functions that are $(n-1)$ times differentiable is given by

$$W[\phi_1, \dots, \phi_n] = \begin{vmatrix} \phi_1 & \phi_2 & \dots & \phi_n \\ \phi_1' & \phi_2' & \dots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n-1)} & \phi_2^{(n-1)} & \dots & \phi_n^{(n-1)} \end{vmatrix}$$

Theorem: If a set of functions $\phi_1, \dots, \phi_n$, each of which has a derivative of order $(n-1)$, is linearly dependent on an interval I , $a \leq t \leq b$, then the Wronskian is zero on I .

Note: This is a necessary condition, not sufficient!

Short version: If dependent, then $W = 0$.

pretty useless for now,
however, contrapositive is not!

Contrapositive: If $W \neq 0$, then independent.

Note: The Wronskian must be zero on the whole domain for the theorem to hold.

Note: What the theorem doesn't say is when $W=0$, then ...?

Nothing! Don't squeeze more from the theorem than it says!

(see book for counterexample, and the upcoming WS for problems.)

Ex// $\{1, \cos(x), \sin(x)\}$
on $[0, 2\pi]$.

$$W = \begin{vmatrix} 1 & \cos(x) & \sin(x) \\ 0 & -\sin(x) & \cos(x) \\ 0 & -\cos(x) & \sin(x) \end{vmatrix}$$
$$= 1 \begin{vmatrix} -\sin & \cos \\ -\cos & \sin \end{vmatrix} = -\sin^2 - (-\cos^2)$$
$$= \cos^2 - \sin^2 = \cos(2x) \neq 0 \text{ on all of } [0, 2\pi].$$

Ex// Chebyshev Polynomials: $T_0(x)=1$ $T_1(x)=x$ $T_2(x)=2x^2-1$, on $[-1, 1]$

$$W[T_0, T_1, T_2] = \begin{vmatrix} 1 & x & 2x^2-1 \\ 0 & 1 & 4x \\ 0 & 0 & 4 \end{vmatrix}$$
$$= (1)(1)(4) = 4 \neq 0 \text{ on } [-1, 1]$$

Two important results.

1) x^n & x^m are independent for any $n \neq m$.

pf:

$$c_1 x^n + c_2 x^m = 0$$

exactly when

$$c_1 + c_2 x^{m-n} = 0 \quad \text{for } x \neq 0.$$

$$\Rightarrow x^{m-n} = -\frac{c_1}{c_2}$$

LHS is a function, RHS is a constant. This is a contradiction except for in the case where $c_1 = c_2 = 0$, in which case

$$c_1 x^n + c_2 x^m = 0 \quad \forall x \in \mathbb{R} \setminus \{0\} \quad \text{and all } n \neq m.$$

2) $\{1, \cos(x), \cos(2x), \dots, \sin(x), \sin(2x), \dots\}$ is a collection of linearly independent vectors over $[0, 2\pi]$. Pf will have to wait for worksheets and later.

1) implies a Taylor Polynomial is an infinite dimensional linear combination of vectors.

2) Similarly, for any integrable function f we can write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

This is called a Fourier Series, absolutely essential in applications and theory!

Span

If a vector space V consists of all linear combinations of the particular vectors $u_1, u_2, \dots, u_k$, then these vectors span the space.

In other words, every vector $v \in V$ can be expressed as some combination of the u 's:

$$v = c_1 u_1 + c_2 u_2 + \dots + c_k u_k \text{ for some coefficients.}$$

Note: We can have many combinations of c_i 's.

Alt. Defn: A span is the collection of all linear combinations of vectors in the given vector space. We write: $\text{span}[u_1, \dots, u_k]$

$$\text{Ex// } \vec{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \vec{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \vec{u}_3 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$$

$\vec{u}_1, \vec{u}_2$ span a 2D space

$\vec{u}_2, \vec{u}_3$ span a 2D space

$\vec{u}_1, \vec{u}_3$ span only a 1D space

Since $2\vec{u}_1 = \vec{u}_3$. They're effectively the same vector.

// From the definition of span, the column space of a matrix is the space spanned by the columns of that matrix.

We denote this space by $\text{Col}(A)$.

Theorem: For a set of vectors $\{u_1, \dots, u_k\}$ in a vector space V , $\text{span}[u_1, \dots, u_k]$ is a subspace of V .

pf: let $v_1 \neq v_2$ be two vectors in the span.

$$v_1 = a_1 u_1 + \dots + a_k u_k$$

$$v_2 = b_1 u_1 + \dots + b_k u_k$$

$$\begin{aligned} \text{then } \alpha v_1 + \beta v_2 &= \alpha(a_1 u_1 + \dots + a_k u_k) \\ &\quad + \beta(b_1 u_1 + \dots + b_k u_k) \\ &= (\alpha a_1 + \beta b_1) u_1 + \dots + (\alpha a_k + \beta b_k) u_k \\ &= c_1 u_1 + \dots + c_k u_k. \checkmark \end{aligned}$$

Ex// Consider $B = \begin{bmatrix} 1 & 3 & 0 & 1 & -2 \\ 2 & 4 & 1 & 1 & 5 \end{bmatrix}$

$\text{Col}(B)$ is the column space of B . There are five vectors that span a 2D space.

Note:

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} -2 \\ 5 \end{pmatrix} = 7 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + (-9) \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

All vectors can be written as combinations of two of them. We chose $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ & $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ as an example. So,

$$A\vec{x} = \vec{b} \Rightarrow c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \vec{b}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 5 \end{pmatrix} \right\}$$

$$= \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} = \mathbb{R}^2$$

Summary thus far:

Linear Independence involves the Null Space "Row Picture"

Span involves the Column Space "column picture"

Both Null Space & Column Space are subspaces of V !

(NOT a coincidence)

Basis

A basis for a vector space is a collection of vectors having two properties at once:

- ① They are linearly independent
- ② They span the space.

This is saying any vector in a space is a linear combination of basis vectors!
(unpack that sentence!)

Ex/ $\hat{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is by itself linearly independent, but it does span $\mathbb{R}^2$.

Ex/ $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ span $\mathbb{R}^2$, but they are not linearly independent since

$$(1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (1)\begin{pmatrix} 0 \\ 1 \end{pmatrix} + (-1)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \vec{0}$$

Ex/ $\begin{pmatrix} 1 & 3 & 0 & 1 & -2 \\ 2 & 4 & 1 & 1 & 5 \end{pmatrix}$. In RREF $\begin{pmatrix} 1 & 0 & 3/2 & -1/2 & 23/2 \\ 0 & 1 & -1/2 & 1/2 & -9/2 \end{pmatrix}$ two pivot columns

The pivot columns are exactly the columns that correspond to vectors that form a basis for that space ($\mathbb{R}^2$)

- There are too many to be linearly independent
- The columns span $\mathbb{R}^2$

A basis is the perfect collection of vectors that span the space and are linearly independent!

The number of these vectors is called the dimension of the space.

Despite the infinite number of basis vectors one can have, the dimension is an intrinsic part of the space!

Thus, in $\mathbb{R}^n$, a basis has at least n vectors that span $\mathbb{R}^n$, and at most n vectors that are linearly independent.

- More than n vectors can span $\mathbb{R}^n$
- Less than n vectors may be linearly independent.
- Any linearly independent set in $\mathbb{V}$ can be extended to a basis by adding vectors if necessary.
- Any spanning set in $\mathbb{V}$ can be reduced to a basis by deleting vectors if necessary.

- A basis is a maximal independent set. Cannot be made larger w/o losing independence.
- A basis is a minimal spanning set. Cannot be made smaller w/o losing span.

$$\text{Ex} // v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad v_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Are these vectors lin. dep.? What space do they span? Do they form a basis?

Note, to tell independence we need $c_1 = c_2 = c_3 = c_4 = 0$ in the equation

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = 0$$

$$\text{or} \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

A few ways to tell:

REF: $\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ so 3 pivot columns means it spans $\mathbb{R}^3$. Not also

$$(1) \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + (1) \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

so not all independent.

Thus the vectors do not form a basis of $\mathbb{R}^4$.

Ex// Consider the same matrix as before:

$$\begin{pmatrix} 1 & 3 & 0 & 1 & -2 \\ 2 & 4 & 1 & 1 & 5 \end{pmatrix}$$

The columns correspond to variables $x_1, x_2, x_3, x_4, \text{ \& } x_5$

The variables $x_1, \text{ \& } x_2$ are basic variables.

The variables $x_3, x_4, \text{ \& } x_5$ are free variables.

The span of the columns as vectors corresponds to linear combinations of columns corresponding to pivot columns.

The size of this is 2. It is the rank. $r=2$. It is also labeled $\mathcal{R}(A)$ for "range". It is the dimension of the column space.

The pivot columns form a basis for the column space.

The size of the null space corresponds to the number of free variables.

This is the "nullity." It is 3. It is also labeled $\mathcal{N}(A)$.

The maximal linearly independent collections of homogeneous solutions to $A\vec{x}=\vec{0}$ form a basis for the null space.

$$\text{Dimension} = n = \text{rank} + \text{nullity} = r + (n-r)$$

Rank-Nullity Theorem (version 1)

Let A be a $m \times n$ matrix. Then $n = \text{rank} + \text{nullity}$.

Updating the Invertible Matrix Theorem

Given an $n \times n$ matrix A , the following are equivalent:

OLD

NEW

- | | |
|--|---|
| ① A is invertible | ① $\det(A) \neq 0$ |
| ② A^T is invertible | ② $\det(A^T) \neq 0$ |
| ③ A is row equivalent to I_n | ③ The columns of A are linearly independent |
| ④ A has n pivot columns | ④ The rows of A are linearly independent |
| ⑤ $A\vec{x} = \vec{0}$ has only the trivial solution | ⑤ The columns of A form a basis for $\mathbb{R}^n$ |
| ⑥ $A\vec{x} = \vec{b}$ has unique solution $\vec{x} = A^{-1}\vec{b}$ | ⑥ Every column of A is a pivot column |
| ⑦ $\text{Rank}(A) = n$ | ⑦ $\dim(\text{Col}(A)) = n$ |
| ⑧ $A\vec{x} = \vec{b}$ is definite | ⑧ The columns of A span $\mathbb{R}^n$ |
| ⑨ $A\vec{x} = \vec{b}$ has n basic variables | ⑨ The rows of A span $\mathbb{R}^n$ |
| ⑩ $A\vec{x} = \vec{b}$ has no free variables | ⑩ $\text{Col}(A) = \mathbb{R}^n$ |
| | ⑪ The null space of A is $\{\vec{0}\}$ |
| | ⑫ The nullity of A is 0 |
| | ⑬ The orthogonal complement of $\text{Col}(A)$ is $\{\vec{0}\}$ |
| | ⑭ The orthogonal complement of the null space is $\mathbb{R}^n$. |